



How definitive treatment affects the rupture rate of unruptured cerebral aneurysms: a competing risk survival analysis

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OBJECTIVE To investigate the risk of bleeding from unruptured cerebral aneurysms (UCAs), previous studies have used Kaplan-Meier analyses without treating the definitive treatment as a competing risk event, which may underestimate the rupture rate. The authors analyzed the survival of patients with UCAs alongside the occurrence of aneurysm bleeding and its competing risk events.

METHODS A retrospective analysis was conducted on 722 patients diagnosed with UCAs in the period from 2000 to 2009 using an institution's electronic medical records and telephone interviews. The cumulative incidence of aneurysm rupture was examined, and factors contributing to rupture were assessed using regression analyses.

RESULTS By 2014, 19 patients had experienced aneurysm rupture, with an overall rupture rate of 0.57% per year over 3320.8 person-years. However, cumulative incidence analysis indicated that 1.3% of all patients had a rupture within 2 years after aneurysm identification, and 38.4% of the patients underwent definitive treatment in the same period. Among the patients who experienced rupture, regression analysis revealed that an aneurysm size greater than 5 mm, a location in the anterior or posterior communicating arteries, and an irregular shape contributed to aneurysm rupture, with HRs of 4.4 (95% CI 1.2–15.7), 3.5 (95% CI 1.4–8.7), and 2.1 (95% CI 0.7–6.0), respectively.

CONCLUSIONS Rupture rate analyses using the person-year or standard Kaplan-Meier method are not as informative without consideration of the competing risks. The incidence of aneurysm rupture should be presented clearly with those of competing risks.

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KEYWORDS competing risk; unruptured cerebral aneurysms; subarachnoid hemorrhage; survival analysis; etiology; vascular disorders

ANEURYSMAL subarachnoid hemorrhage (aSAH) continues to be associated with high morbidity and death despite the refinement of treatment modalities. Rupture rates and risk factors associated with bleeding from unruptured cerebral aneurysms (UCAs) have been reported in many retrospective and several prospective studies using Kaplan-Meier curves and Cox proportional-hazards analyses.^{2–5,7–10,12,13,16,19–23,25,26} However, Kaplan-Meier survival analysis is known to be less informative in the

presence of competing risks,⁶ which should be discussed in the clinical course of a UCA. Because aSAH can be lethal or cause severe morbidity, leaving all UCAs untreated and monitoring their natural course is ethically challenging. Hence, aneurysms with considerable risk factors for rupture, including larger aneurysms or lesions in patients with a history of aSAH, should be treated with clipping or endovascular embolization. Because treated aneurysms do not bleed, definitive treatment is an important competing

ABBREVIATIONS ACoA = anterior communicating artery; aSAH = aneurysmal subarachnoid hemorrhage; BA = basilar artery; ICA = internal carotid artery; MCA = middle cerebral artery; PCoA = posterior communicating artery; UCA = unruptured cerebral aneurysm; VA = vertebral artery.

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risk associated with aneurysm bleeding, along with death from causes other than aSAH. In addition, because definitive treatment is not performed randomly but is geared toward patients with a considerable risk of developing aSAH, with considerable life expectancy, and with manageable treatment risks, it is “not a ‘non-informative’ or ‘random’ occurrence, which the Kaplan-Meier estimator regards as the basis for a censoring event.”⁶ In scenarios with relevant competing risks, it is important to recognize the cumulative incidence of each competing risk event when interpreting the analysis.¹⁵ Here, we searched for patients with UCAs in our institution’s electronic health records and investigated the current status of these patients. The rupture rate of UCAs and the presence of competing risk events are presented.

Methods

This investigation was a single-institution retrospective cohort study. Electronic medical records were introduced at our institution in December 2000. Since that time, all medical records, including radiological images, have been continuously accumulated on the server. We searched these electronic medical records using diagnostic terms related to UCAs and aneurysms, including those in large vessels. The records of 4145 patients were tagged with these diagnostic terms. A review of these patients’ medical records and radiological images revealed that 907 of these patients had cerebral aneurysms. Among them, 106 patients received consultation for a second opinion alone and had no radiological data on the server and/or did not have any subsequent follow-up visits to our institution. After the exclusion of patients with cavernous sinus aneurysms, fusiform aneurysms, or flow-related aneurysms accompanied by arteriovenous malformation, 722 patients with intradural saccular aneurysms were included in this study.

Data on aneurysm characteristics, known risk factors for bleeding, and aneurysm status (untreated, treated, or bled) were collected from each patient’s medical record and via telephone interviews. Aneurysms were confirmed on the DICOM viewer, and their size was recorded as the maximum diameter using maximum intensity projection (MIP) or the original CT angiography or MR angiography images. If the aneurysm showed an irregular protrusion of the wall or a multilobular shape in the MIP image, it was defined as irregularly shaped. Interobserver variability was not investigated in this study.

For survival analyses, the survival period was calculated beginning from the date of the oldest imported radiological study indicating the relevant aneurysm and ending at the earliest of the following: 1) treatment date of the last aneurysm; 2) aSAH onset date; 3) death date; 4) December 31, 2014; or 5) last hospital visit day for patients lost to follow-up. For aSAH, diagnoses by radiological examination or cerebrospinal fluid analysis were included. However, cases involving sudden death or death without an autopsy were excluded from the aSAH cases.

Overall rupture rate was calculated using the person-year method.

To clearly indicate each competing risk event, the cu-

mulative incidence of each event was plotted with the population at risk. A standard Kaplan-Meier curve was also made for reference. In addition, regression analysis was performed to analyze the factors contributing to aneurysm bleeding considering the competing risks.

Since the study was completed before the first author moved to another hospital, our institution’s (NTT Medical Center Tokyo) ethics board approved this investigation after the first author transferred to his present hospital. This study was conducted according to the STROBE statement.

Statistical Analysis

IBM SPSS Statistics version 20 for Windows (IBM Corp.) was used for descriptive statistics and for depicting the Kaplan-Meier curve. Categorical variables were compared using chi-square tests. Continuous variables were compared among the groups using the Mann-Whitney U-test or Student t-test. R version 3.1.0 (The R Foundation for Statistical Computing, <http://www.R-project.org/>) and two statistical packages^{17,18} were used for regression analysis and the plotting of cumulative incidence curves. The threshold used for statistical significance was $p < 0.05$.

For regression analysis, the aneurysm size was dichotomized into smaller than 5 mm and 5 mm or larger because aneurysms smaller than 5 mm were usually observed without surgical intervention.^{11,19} Aneurysm location was examined after categorization into 7 regions, i.e., anterior communicating artery (ACoA), middle cerebral artery (MCA), internal carotid artery–posterior communicating artery (ICA-PCoA) bifurcation, distal to basilar artery (BA), proximal to vertebral artery (VA) union, ICA other than the ICA-PCoA, and others. Several combinations of these regions were examined as covariates, and the most appropriate regression model was selected according to Bayesian information criteria.^{17,18}

Results

Patient characteristics are listed in Table 1. Surgical intervention occurred in 336 cases (313 open surgeries and 23 endovascular treatments).

Nineteen patients experienced radiologically confirmed aSAH; of those, 7 died due to aSAH. During the same period, 37 patients died from causes other than aSAH, such as cancer or heart disease. In addition, 5 other patients experienced sudden death or were found dead but were not subsequently assessed via radiological investigation or autopsy. Seventeen patients were lost to follow-up.

The total observation period was 3320.8 person-years (average 4.6 years, median 4.9 years, IQR 7.6 years), and the overall rupture rate by the person-year method was 0.57% per year.

Cumulative Incidence Curves

The cumulative incidences of aneurysm rupture, definitive treatment, and death from causes other than aSAH are depicted in Fig. 1. While 1.3% of all patients had aSAH within 2 years after aneurysm identification, 38.4% underwent definitive treatment in the same period. In addition, a standard Kaplan-Meier curve was made for reference (Fig. 2). Definitive treatment and death from causes other than

TABLE 1. Summary of characteristics in 722 patients with intradural saccular aneurysms

Factor	No. (%)	Treatment		p Value
		Yes	No	
Age (yrs)		60.3 ± 9.7	64.8 ± 11.3	<0.001
Female	463 (64)	221	242	0.434
History of SAH	16 (2)	7	9	1
Family history of SAH	93 (13)	48	45	0.347
Smoking	280 (39)	138	142	0.271
Alcohol consumption	301 (42)	150	151	0.154
Treatment of hypertension	338 (47)	157	181	1
Treatment of dyslipidemia	248 (34)	108	140	0.277
Size of index aneurysm in mm		7.1 ± 5.1	4.7 ± 3.3	<0.001
≥5 mm	356 (49)	233	123	<0.001
≤3 mm	201	38	163	
>3 & <5 mm	165	65	100	
≥5 & <7 mm	156	97	59	
≥7 & <15 mm	170	115	55	
≥15 & <25 mm	23	16	7	
≥25 mm	7	5	2	
Bleb	326 (45)	191	135	<0.001
Multiple aneurysms	126 (17)	71	55	0.02
Lesion location				
ACoA/PCoA	222 (31)	130	92	<0.001
MCA	263	131	132	
ICA-PCoA	101	61	40	
ACoA	122	70	52	
ICA	166	44	122	
Distal to BA	32	8	24	
VA	16	8	8	
Other	22	14	8	

ICA-PCoA = ICA-PCoA bifurcation; ICA = ICA other than ICA-PCoA; VA = proximal to VA union.

aSAH are depicted as short vertical lines (censored), as are loss to follow-up and end of the study period.

Regression Analysis

Factors contributing to aneurysm rupture were analyzed using the multivariate regression analysis proposed by Fine and Gray.¹ The regression model best explaining contributions to aneurysm bleeding included the following factors: aneurysm size greater than 5 mm, location of the aneurysm in the ACoA or PCoA, and irregular aneurysm shape (Table 2). Aneurysm size larger than 5 mm and aneurysm presence on the ACoA or PCoA showed HRs of 4.36 (95% CI 1.21–15.77) and 3.48 (95% CI 1.39–8.72), respectively. Irregular aneurysm shape was not significant but was associated with a tendency to bleed (HR 2.09, 95% CI 0.73–5.98) and was incorporated into the most appropriate regression model in this analysis after comparison of Bayesian information criteria.

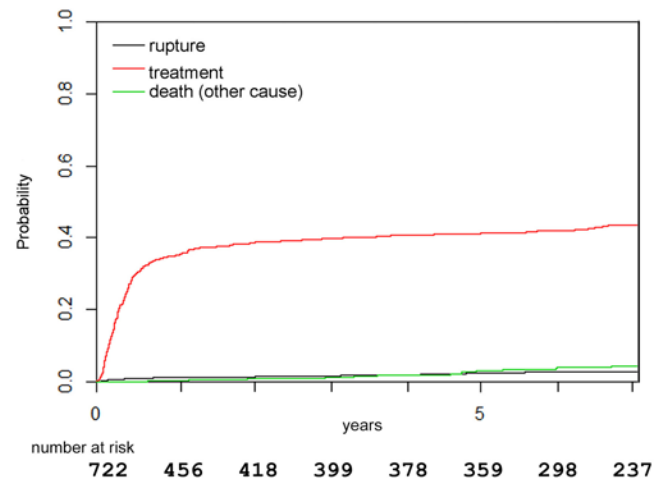


FIG. 1. Cumulative incidence of each event: rupture, treatment, death. Note that 38.4% of patients received definitive treatment within 2 years, and 1.3% of these cases had rupture during this same period.

Discussion

In this study, we evaluated the clinical course of UCAs by retrospectively reviewing our hospital's electronic medical records. There were 19 ruptures among 3320.8 person-years, which equated to a rupture rate of 0.57% per year. While 1.3% of the patients developed aSAH within 2 years after aneurysm discovery, 38.4% underwent definitive treatment in the same period. An aneurysm size greater than 5 mm, aneurysms in the ACoA or PCoA, and an irregular aneurysm shape were incorporated into the regression model of factors contributing to aneurysm rupture.

Since Wiebers' seminal study,²⁴ many studies have described the natural history of aneurysms, and the risk factors associated with SAH have been investigated using Cox proportional regression analysis and Kaplan-Meier estimation. However, because bleeding from an aneurysm can be life threatening, observing all aneurysms without treatment is ethically challenging. Instead, aneurysm patients with known risk factors such as a large aneurysm size or a history of aSAH tend to be treated with surgical clipping or endovascular treatment. Because these definitive treatments and death from other causes are events that prevent aneurysms from bleeding, they must be considered "competing risks" for aneurysm bleeding.

The Kaplan-Meier method includes censoring for random events or events independent of a patient's characteristics and regards the censored object as continuing to have the same risk as if it were not censored. In the presence of competing risks, Kaplan-Meier estimation is generally considered less informative.⁶ When objects considered to be more susceptible to a certain event, such as large aneurysms, are selectively "censored" as if the censoring were independent of the patient characteristics, the risk of the event is underestimated.¹⁴ In the context of the clinical course of UCAs, larger lesions would be more prone to definitive treatment, as previous reports have shown that aneurysm size is the primary risk factor for aSAH. In the same manner, neurosurgeons do not treat UCAs randomly but preferentially treat aneurysm patients with bleeding

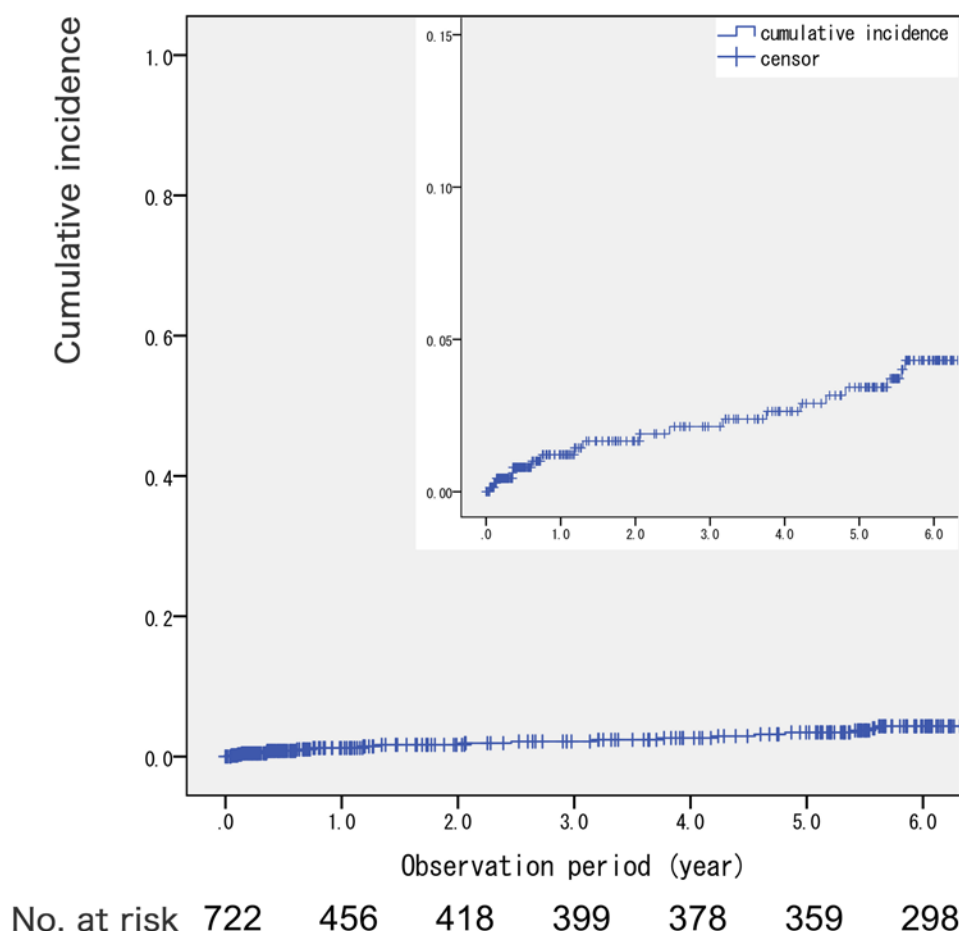


FIG. 2. Standard Kaplan-Meier curve handling definitive treatment and death from causes other than aSAH as censored, as well as loss to follow-up and end of the study period. **Inset:** The same data on an enlarged y-axis.

risks that cannot be ignored based on previous reports, those with a considerable life expectancy, and those with acceptable risks associated with treatment complications.

The treatment indication for cerebral aneurysms is not determined randomly, whereas the Kaplan-Meier method assumes that the censoring event occurs randomly and should be independent of the patient characteristics. When definitive treatment is handled as a censored and random event, an underestimation of the rupture rate can occur (1 – survival probability). Thus, Kaplan-Meier estimation inevitably leads to bias when evaluating the risk of UCAs or the associated rupture rate. Juvela and colleagues^{4,5} reported the natural history of UCAs without surgical intervention. Although death from causes other than aSAH cannot be ignored as a competing risk in the long run, the authors' results are still important in judging the surgical indication for UCAs because a major competing risk—definitive treatment—was eliminated. The Small Unruptured Intracranial Aneurysm Verification (SUAVE) study,¹⁹ in which aneurysms smaller than 5 mm were observed without definitive treatment, could have generated another unbiased analysis of the UCA rupture risk. However, other studies, including ours, have significant biases in determining rupture rates, as a large proportion of patients with aneu-

rysms undergo definitive treatment shortly after inclusion (Fig. 1). For example, the Unruptured Cerebral Aneurysm Study of Japan (UCAS Japan) showed that one-third of aneurysm patients underwent definitive treatment in the first 3 months following aneurysm registration.¹¹

The overall rupture rate of our cohort calculated using the person-year method was very low at 0.57% per year. As shown in Fig. 1, most definitive treatments were performed in the early observation period, presumably for the patients who had a considerable risk of aSAH. In addition, the observation period for low-risk aneurysms was extensive, leading to the low rupture rate as calculated by event/(person × observation years).

TABLE 2. Factors contributing to aneurysm rupture

Factor	HR	95% CI		p Value
		Lower Limit	Upper Limit	
Size larger than 5 mm	4.36	1.21	15.77	0.025
ACoA/PCoA	3.48	1.39	8.72	0.0076
Irregular shape	2.09	0.73	5.98	0.17

Boldface type indicates statistical significance.

In clinical practice, information on UCA rupture rates affects patient decisions on whether to undergo definitive treatment, as this treatment is preventive. The indication for definitive treatment is influenced by the life expectancy of the patient and the calculated cumulative risk of aSAH based on annual rupture rates. Therefore, these biases should be considered carefully and presented clearly.

In the presence of competing risks, visual presentation of each competing risk event is considered an effective approach to show possible biases.¹⁵ For analyzing the risk of aSAH, the percentage of patients who undergo definitive treatment should be presented, as this accounts for a large fraction of the cohort,¹¹ and patients in the cohort were considered at a sufficiently high risk for aSAH to warrant the treatment.

Gray and Fine proposed direct regression modeling of the effect of covariates on the cumulative incidence function for “each” competing event,¹ and it is regarded as the hazard of failure from a given cause under the existence of competing events. According to our results, aneurysms larger than 5 mm, aneurysms in the ACoA or PCoA, and an irregular aneurysm shape were included in the most appropriate regression model for aneurysm bleeding in the presence of competing risks. These risk factors were found to be similar to those for the UCAs in the UCAS Japan.¹¹ However, our data indicate that the regression model including these three factors or covariates most appropriately explains the contributions to aSAH under treatment indication criteria during the observation period. Our results suggest that to further reduce the occurrence of aSAH, aneurysms with these three factors should be treated more intensively in addition to the treatment indication of that period.

In our cohort, smoking, posterior circulation aneurysms, and family history were not incorporated into the most appropriate regression model to describe their contribution to aSAH for two reasons. First, the number of documented aSAH cases was so small that contributing factors other than aneurysm size could be strongly affected by the characteristics of the aSAH patients. Alternatively, characteristics of the ruptured aneurysm may have been overestimated because of the small number. Second, regarding factors that were reported as risks for aSAH in the study period, the patients with such factors may have been more prone to undergo definitive treatment than those without, or smoking cessation may have been strongly encouraged. This tendency would also affect future studies because patients with known risk factors with large hazard ratios in previous studies would be prone to undergo treatment, which should modify the true risk factor.

Our study has several limitations, such as its retrospective single-institution design. Moreover, the 5 recorded sudden deaths may have occurred because of aSAH, perhaps leading to a further underestimation of the rupture rates. As described above, the small number of documented aSAH cases may have resulted in hazard ratios with wide confidence intervals and may have led to the misvaluation of other risk factors. In the regression analysis, we assumed that the proportional hazards were the same as those in previous studies, which should be more precisely confirmed. In fact, some aneurysms can grow

in size or show morphological changes, and a smoking habit and other medical factors, such as hypertension, can be modified over time, which disrupts the proportional hazard assumption. In this analysis, however, we focused on the existence of competing risks, especially definitive treatment, which occurred much more often than bleeding from aneurysms and which should be a strong bias. Because the majority of definitive treatments occurred early in the observation period, the effect due to deviation from the proportional hazard assumption would be small.

Conclusions

While 1.3% of our cohort cumulatively experienced radiologically confirmed aSAH in the first 2 years, 38.4% of the population underwent definitive treatment. Within the aSAH group, a regression model including an aneurysm size larger than 5 mm, an ACoA or PCoA aneurysm location, and an irregular shape was most appropriate in explaining the contributions to bleeding. Because definitive treatment is a major competing risk against the rupture of UCAs and it usually occurs in the early period of the survival analysis, it contributes to an underestimation of the rupture rate of UCAs.

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Disclosures

The authors report no conflict of interest concerning the materials or methods used in this study or the findings specified in this paper.

Author Contributions

Conception and design: Kimura, Kawai. Acquisition of data: Kimura. Analysis and interpretation of data: Kimura. Drafting the article: Kimura. Critically revising the article: Ochiai, Kawai, Morita. Reviewed submitted version of manuscript: Morita. Approved the final version of the manuscript on behalf of all authors: Kimura. Statistical analysis: Kimura. Study supervision: Kawai, Saito.

Supplemental Information

Previous Presentations

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